$$\bar{\mathbf{A}} = \mathbf{A} - \langle \mathbf{A} \rangle$$

$$\bar{\mathbf{B}} = \mathbf{B} - \langle \mathbf{B} \rangle.$$

Then repeat the whole process, replacing **A** and **B** with $\bar{\mathbf{A}}$ and $\bar{\mathbf{B}}$. The following exercise serves as a guide.

Exercise 5.2:

1) Show that $\Delta A^2 = \langle \bar{\mathbf{A}}^2 \rangle$ and $\Delta B^2 = \langle \bar{\mathbf{B}}^2 \rangle$.

2) Show that $[\bar{\mathbf{A}}, \bar{\mathbf{B}}] = [A, B]$.

3) Using these relations, show that

$$\Delta \mathbf{A}\ \Delta \mathbf{B} \geq \tfrac{1}{2} |\langle \Psi | [\mathbf{A}, \mathbf{B}] | \Psi \rangle|.$$

Later, in Lecture 8, we will use this very general version of the uncertainty principle to prove the original form of Heisenberg's Uncertainty Principle: *The product of the uncertainties of the position and momentum of a particle cannot be less than half of Planck's constant*.